

INDIVIDUAL PROJECT SUMMARY REPORT

Calibo AI Academy | Phase 1, Mini Use Case 01 | Category Intelligence | Path 2, KLU Batch

Student Name	GUDA ABHINAV SAI
Team	Team 07
Project Title	Category Intelligence — Retail Product Performance EDA
Individual Role	Business Synthesiser
Sections Owned	Step 5 (Time Patterns) · Step 6 (Category Manager Briefing)
Tools Used	Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook
Dataset	MUC01_Retail_Sales_Dataset.csv — 107,836 transactions, 12 stores across 6 Andhra Pradesh cities, Jan–Jun 2026

CBIM Problem Canvas

Business Problem	The chain runs promotional spend without knowing when demand is naturally strong, and the Category Manager must walk into a supplier review able to state a position on all five categories rather than present a set of charts.
Business Objective	Establish when demand occurs across the week and the half-year, and synthesise every team finding into a briefing the Category Manager can act on without asking a follow-up analytical question.
Data Used	MUC01_Retail_Sales_Dataset.csv, as validated in Step 1, together with the findings produced in Steps 2, 3 and 4 by the Data Analyst and Insights Lead.
Key Analysis Focus	Step 5 — Day-of-week and monthly demand patterns, chain-wide and for the focus category. Step 6 — The Category Manager Briefing, plus the limitations governing it.
Expected Business Output	A recommendation on promotional timing, and a briefing naming the category to back, the supplier to challenge with a figure, the discount verdict, and the data still missing.

1. Project Overview

The project analyses six months of retail sales data for a chain of 12 stores, to produce an evidence pack for a quarterly supplier review. The work is divided across six steps and three roles. My role as Business Synthesiser covers the closing stage: establishing when demand actually occurs, and converting the team's separate findings into a single briefing the Category Manager can use. Where the earlier steps each answer one question, my sections must make those answers consistent with one another and honest about their limits.

2. My Individual Contribution — Business Synthesiser

Time pattern analysis. I computed average daily revenue by day of week and total revenue by month, identified the best and worst trading day and month for the chain, and repeated the analysis for Electronics — the category identified in Step 2 as the most consequential. To compare categories of very different sizes I indexed each to its own weekly average, which isolates the *shape* of the week from its scale. I then drew the promotional timing conclusion and stated the qualification that limits it.

Synthesis and briefing. I wrote the Category Manager Briefing, drawing on the category trend and decomposition from Step 2, the product finding from Step 3 and the discount verdict from Step 4, ensuring every claim carried a category name, a figure and an action. I also compiled the limitations section governing the whole analysis, including testing the seasonality limitation rather than merely asserting it.

3. Step 5 — Day-of-Week and Monthly Patterns

Purpose. Establish when demand is naturally strong and when promotional spend is being directed into an empty shop.

Method note. "Average daily revenue by day" means totalling revenue for each calendar date and then averaging those daily totals within each weekday. Averaging transaction-level revenue instead would give average basket size — a different question with a different answer, and a common error. I used the first method throughout.

Findings. Chain-wide, Saturday is the best trading day at Rs 3.52M average and Monday the worst at Rs 2.01M, a gap of 75.1%. Friday through Sunday carries the week. January is the best month at Rs 84.8M and June the worst at Rs 69.3M, consistent with the decline established in Step 2.

Electronics shares the chain's best and worst day and month, but not its amplitude. Indexed to its own average day, Electronics runs at 142.8 on Saturday against 77.6 on Monday — a 65-point swing — while Apparel is visibly flatter at 131.5 and 82.0. So the focus category follows the same pattern as the chain while being markedly more extreme.

Business interpretation. A high-consideration purchase such as a laptop or phone requires an unhurried visit, so Electronics demand concentrates into the weekend, while Apparel picks up incidental weekday footfall. Promotional spend should therefore be concentrated Friday to Sunday and pulled out of Monday and Tuesday, where no category shows latent demand to unlock. This applies most strongly to Electronics: the category most in need of recovery is also the one whose customers are least available midweek.

Qualification stated. Concentrating promotions on Saturday is demand *capture*, not demand *generation* — it discounts customers who were already coming. Given that Step 2 identified the problem as 33.4% fewer transactions, and Step 4 showed discount depth generates no volume, timing is the cheaper lever but it is not the cure.

4. Step 6 — Category Manager Briefing

Purpose. Convert five separate analyses into a position the Category Manager can state out loud in the supplier review.

The briefing. Apparel has the strongest case for increased investment: it grew 34.1% on 31.3% more transactions with price flat, and is the only category where all five products grew. The Electronics supplier should be challenged, specifically on Mobile Phone — a 57.1% decline representing Rs 16.0M of lost monthly run-rate from a category holding 49.3% of chain revenue, caused by 33.4% fewer transactions while our prices moved only 2.2%. On discounts, Rs 35.6M of revenue was given away across 66.6% of transactions with volume lift statistically ruled out above 1.4%, so promotional support should not be renewed at current depth without a committed volume target. The single piece of data still needed before a final decision is cost and margin per product.

Limitations compiled. Everything above is a revenue finding; the dataset contains no cost column, so the brief's question about "eroding margin" cannot be answered and the reallocation is a well-evidenced hypothesis rather than a proven allocation. The window also misses the Indian festival season — Dussehra and Diwali are the peak selling period for Electronics and Apparel, and Jan–Jun sits in the trough of that calendar.

Rather than assert that limitation, I tested it. Sankranti, the major Andhra Pradesh festival, falls 14–17 January, yet Electronics that week averaged Rs 1.39M per day against Rs 1.50M for the rest of the month, and all five of January's highest-revenue days were Saturdays. The dataset contains a weekly cycle and a downward trend and no seasonal structure at all, so the trend cannot be projected into the festival quarter. This changes the recommendation rather than merely qualifying it: the Electronics finding should be treated as a supplier conversation, not a shelf-space cut, until at least one prior festival season has been observed.

5. Skills and Concepts Applied

Datetime feature extraction; distinguishing between aggregating by calendar date and by transaction, and understanding why the two answer different questions; indexing to a category's own mean to compare shape independently of scale; comparative pattern analysis across subgroups; hypothesis testing against a stated limitation; and synthesis — reconciling findings from separate analyses into one consistent narrative with figures that agree across every deliverable.

6. Learning Outcomes

The clearest lesson was that how a metric is computed can change its answer entirely. Averaging transaction revenue by weekday makes the week look almost flat and identifies the wrong worst day; totalling by date first reveals a 75% spread. The definition mattered more than the code.

I also learned that a good synthesis states what weakens its own conclusion. Noting that weekend promotion is demand capture rather than generation, and that the festival season is absent from this window, makes the briefing more credible rather than less — and the festival point in particular changed what we recommend the Category Manager actually do. Testing that limitation against the data, rather than simply claiming it, was the more valuable step.

7. Conclusion

My contribution closed the analysis. Step 5 established that the chain's week is heavily back-loaded and that Electronics is the most weekend-dependent category, producing a promotional timing recommendation together with an honest statement of what timing alone cannot fix. Step 6 synthesised the team's five analyses into a briefing naming the category to back, the supplier to challenge with a rupee figure, the discount position and the data still missing — and compiled the limitations, including a tested rather than assumed account of why this six-month window cannot be projected into the festival quarter.

All figures in this report are reproducible from the team notebook, Steps 5 and 6. Currency figures are in Indian Rupees. This analysis covers revenue only; the dataset contains no cost column, so profitability cannot be assessed.